A SUBMODULE GENERATING A FINITE FREE MODULE OVER A SEMI-LOCAL RING CONTAINS A BASIS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted proof refers to Nakayama's lemma by label; it is recorded as a navigation stub pointed at its \label line, and the argument consuming it stays inside the counted span.

Editorial note. The counted spans contain no \cite, so the card carries no bibliography.

Allowed standard prerequisites (editorial).

- Nakayama's lemma. Invoked in the counted proof at line 19545. Reduces the search for a basis to a generating condition modulo the Jacobson radical.
- Linear combinations over an infinite field. Used in the counted proof at lines 19556-19558. Produces a single element of the submodule with a nonzero component in every local factor.

Source reference. Original lemma 00DV: sources/stacks-algebra/pinned-original-source.tex, source lines 3833–3833. Navigation pointer for \ref{lemma-NAK} used inside the counted span (line 19545); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a local ring with maximal ideal $\mathfrak{m}$ and infinite residue field. Let $R \to S$ be a ring map. Let M be an S-module and let $N \subset M$ be an R-submodule. Assume*

(1) *S is semi-local and $\mathfrak{m}S$ is contained in the Jacobson radical of S,*
(2) *M is a finite free S-module, and*

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(3) *N generates M as an S-module.*

Then N contains an S-basis of M.

Proof. Assume M is free of rank n. Let $I \subset S$ be the Jacobson radical. By Nakayama's Lemma 00DV a sequence of elements $m_1, \ldots, m_n$ is a basis for M if and only if $\overline{m}_i \in M/IM$ generate M/IM. Hence we may replace M by M/IM, N by $N/(N \cap IM)$, R by $R/\mathfrak{m}$, and S by S/IS. In this case we see that S is a finite product of fields $S = k_1 \times \ldots \times k_r$ and $M = k_1^{\oplus n} \times \ldots \times k_r^{\oplus n}$. The fact that $N \subset M$ generates M as an S-module means that there exist $x_j \in N$ such that a linear combination $\sum a_j x_j$ with $a_j \in S$ has a nonzero component in each factor $k_i^{\oplus n}$. Because $R = k$ is an infinite field, this means that also some linear combination $y = \sum c_j x_j$ with $c_j \in k$ has a nonzero component in each factor. Hence $y \in N$ generates a free direct summand $Sy \subset M$. By induction on n the result holds for M/Sy and the submodule $\overline{N} = N/(N \cap Sy)$. In other words there exist $\overline{y}_2, \ldots, \overline{y}_n$ in $\overline{N}$ which (freely) generate M/Sy. Then $y, y_2, \ldots, y_n$ (freely) generate M and we win. $\square$